

Notes: Blader, Gartenberg & Prat (RES, 2020)

The Contingent Effect of Management Practices

Contents

1	Big picture	3
2	Incremental contribution	3
2.1	Contribution relative to the management-practices literature	3
2.2	Contribution relative to culture and relational-contract theories	3
2.3	Contribution relative to incentives and relative performance feedback	4
3	Data and measurement	4
3.1	Firm setting and EOBR technology	4
3.2	Phase 1 of the lean-management initiative	4
3.3	Performance posting treatments	5
3.4	Outcome variables	5
3.5	Randomization and stratification	5
4	Conceptual model	5
4.1	Direct job satisfaction without collective values	5
4.2	Collectivist values and overall utility	6
4.3	Proposition and economic logic	6
5	Empirical specifications	7
5.1	Pooled intent-to-treat effect	7
5.2	Triple-difference specification	7
5.3	Selection diagnostics	7
5.4	Mechanism tests	7
6	Identification and empirical design	8
6.1	What is randomized	8
6.2	What is not randomized	8
6.3	Why the design is informative	8
6.4	Limitations	8
7	Tables and figures	9
7.1	Figure 1: experimental timeline	9
7.2	Table 1 and Table 2: balance	9
7.3	Figure 2 and Figure 3: preliminary event-study evidence	10
7.4	Table 3 and Table 4: main intent-to-treat estimates	11
7.5	Table 5 and Table 6: selection robustness	12
7.6	Table 7: heterogeneity by pre-posting performance quartile	12
7.7	Table 8 and Figure 4: engagement survey mechanism	13
8	Short summary	14

9	Conclusion	14
9.1	Core conclusion	14
9.2	Economic intuition	15
9.3	Takeaway	15

1 Big picture

- **Question.** Does the effectiveness of a management practice depend on the values and culture articulated by management?
- **Core setting.** The authors run a field experiment inside a large U.S. transportation company during the rollout of electronic on-board recorders (EOBR) for trucks.
- **Management practice.** The focal practice is public performance posting: drivers' EOBR-based driving efficiency metrics are posted at the worksite.
- **Values intervention.** The firm is also rolling out Phase 1 of a large lean-management transformation. Phase 1 emphasizes teamwork, empowerment, democratic management style, and collective problem solving. It changes espoused values rather than driver compensation or formal operating rules.
- **Treatment design.** Sites are assigned to one of three performance-information conditions: control, named public posting, or anonymous-ID posting. Assignment is stratified by Phase 1 status and EOBR rollout week.
- **Main result.** Named public posting improves driver performance in sites that have not yet received Phase 1, but worsens performance in sites that have received Phase 1.
- **Mechanism.** The negative effect in Phase 1 sites is consistent with a conflict between an individualistic, competition-based management practice and a collectivist value system.
- **Important nuance.** Anonymous-ID posting does not generate the same strong negative response. The backlash appears to require that drivers can identify who is outperforming whom.
- **Mechanism evidence.** Phase 1 sites have more collectivist survey responses, and replacing the Phase 1 dummy with a survey-based collectivist index reproduces the triple-difference pattern.
- **Takeaway.** The same formal management practice can have opposite effects depending on the intangible values environment in which it is embedded.

2 Incremental contribution

2.1 Contribution relative to the management-practices literature

The paper contributes to the management-practices literature by showing that the effectiveness of a practice is not universal. Prior work emphasizes that certain practices, such as monitoring, feedback, targets, and performance incentives, are associated with productivity. This paper shows that a practice that works in one organizational environment can fail in another.

- The relevant object is not merely whether a practice is adopted.
- The relevant object is whether the practice is complementary with the organization's values.
- The evidence is experimental for the posting treatment, which is unusually strong for a study of management practices inside firms.

Interpretation: The paper moves from asking "what practices are good?" to asking "under what cultural conditions is a practice good?"

2.2 Contribution relative to culture and relational-contract theories

The paper gives field evidence for the idea that intangible organizational factors matter. Theories of relational contracts, trust, and culture argue that formal practices interact with informal beliefs and norms, but direct evidence is difficult because culture is hard to observe and practices are rarely randomized.

- Phase 1 changes espoused values toward teamwork and empowerment.
- The formal posting treatment is randomized.
- The interaction between these two objects identifies a complementarity, or negative complementarity, between formal practices and values.

Interpretation: The study provides empirical discipline for theories in which organizational culture is not just a background condition but an input into the productivity of formal managerial tools.

2.3 Contribution relative to incentives and relative performance feedback

A large literature studies relative performance feedback and tournaments. This paper shows that relative performance information can be harmful when the work environment emphasizes collective identity and peer cooperation.

- Named postings create explicit interpersonal comparison.
- IDed postings preserve some relative information but reduce identifiability.
- The strongest negative effects occur for named postings in Phase 1 sites.

Interpretation: The incremental lesson is that relative-performance schemes should be evaluated not only by their incentive properties but also by the social meaning they acquire within the organization.

3 Data and measurement

3.1 Firm setting and EOBR technology

- The field site is a large U.S. transportation company with many geographically dispersed sites performing similar operating functions.
- The company installs electronic on-board recorders on trucks. These devices record driving behaviour and generate route-specific measures of driving efficiency.
- The EOBR rollout provides new performance data that can be shared with drivers through different information policies.
- The experiment runs from 3 September 2013 to 13 April 2014, during the EOBR implementation period.

Interpretation: The EOBR rollout creates the technological precondition for monitoring and feedback. Without EOBR data, the performance-posting intervention would not be feasible.

3.2 Phase 1 of the lean-management initiative

- The company is engaged in a multi-year lean-management transformation inspired by the Toyota Production System.
- The first phase, Phase 1, focuses on values and workplace relationships rather than formal productivity tools.
- Phase 1 emphasizes teamwork, empowerment, democratic management style, employee recognition, and greater involvement of front-line workers.
- The later phases of the lean transformation, which would change formal operational tools, had not begun during the experiment.
- About 35% of sites had already received Phase 1 at least three months before the experiment began.

Interpretation: Phase 1 is interpreted as a values intervention. This matters because the paper’s core claim is about the interaction between formal practices and espoused values.

3.3 Performance posting treatments

The experiment includes three posting conditions:

1. **Control:** no public posting of individual driver performance.
2. **Treatment Group 1, Named:** driver names are posted next to individual performance metrics.
3. **Treatment Group 2, IDed:** employee IDs are posted next to performance metrics. Drivers can find their own score, but peers cannot easily identify other drivers.

Interpretation: The named and IDed treatments separate relative performance information from social identifiability. This distinction is crucial for testing whether competition and peer comparison drive the treatment effects.

3.4 Outcome variables

The EOBR system provides four main performance outcomes:

- **Miles per gallon:** fuel efficiency measure.
- **Gap score:** difference between potential miles per gallon and actual miles per gallon for a route.
- **Shift score:** measure of efficient shifting events.
- **Excess idle time / fuel lost:** idling-related inefficiency.

The main regression outcome is a driving efficiency index built from EOBR measures. The paper aligns signs so that lower raw scores represent better performance for some variables, and then uses standardized measures in regressions.

Interpretation: The outcomes are objective operational performance metrics rather than survey outcomes. This reduces concern that the main effect is purely perceptual.

3.5 Randomization and stratification

- Posting treatments are randomized across sites.
- Randomization is stratified by EOBR rollout week and Phase 1 status.
- Phase 1 itself is not randomized by the researchers; it follows the company’s rollout plan.
- The authors argue that the Phase 1 rollout is plausibly unrelated to expected treatment effects because it was scheduled around managerial travel and implementation logistics.

Interpretation: The posting treatment is experimentally assigned, while Phase 1 is quasi-experimental. Therefore the paper devotes substantial effort to balance, matching, and selection diagnostics.

4 Conceptual model

4.1 Direct job satisfaction without collective values

Let y_i be worker i ’s performance and let $\bar{y}_{-i}$ be average performance of the other workers in the team:

$$\bar{y}_{-i} = \frac{\sum_{j \neq i} y_j}{n - 1}. \quad (1)$$

Direct job satisfaction is:

$$u_i = y_i + b\bar{y}_{-i} + \rho(y_i - \bar{y}_{-i}) - \frac{1}{2}c_i y_i^2. \quad (2)$$

- y_i captures direct utility from own performance.
- $b\bar{y}_{-i}$ captures direct value from teammates' performance.
- $\rho(y_i - \bar{y}_{-i})$ captures relative performance concerns made salient by posting.
- $\frac{1}{2}c_i y_i^2$ is the convex cost of effort.

Interpretation: Without collectivist values, posting increases the salience of outperforming peers. This pushes drivers toward higher individual performance.

4.2 Collectivist values and overall utility

Phase 1 is modelled as increasing the importance of the reference group. Overall utility is:

$$U_i = (1 - \lambda)u_i + \lambda\bar{u}_{-i}, \quad (3)$$

where

$$\bar{u}_{-i} = \frac{\sum_{j \neq i} u_j}{n - 1}. \quad (4)$$

The parameter λ measures the internalization of collectivist values:

- $\lambda = 0$: purely individualistic evaluation.
- $\lambda = 1$: full weight on teammates' utility.
- Phase 1 is interpreted as an increase in λ .

Interpretation: Phase 1 changes the social meaning of high relative performance. A top performer may now care that outperforming peers publicly imposes psychological or relational costs on teammates.

4.3 Proposition and economic logic

The model yields four key implications:

1. Pre-Phase 1, postings increase performance.
2. There is a negative complementarity between Phase 1 and postings:

$$\frac{\partial^2 \hat{y}_i}{\partial \lambda \partial \rho} < 0. \quad (5)$$

3. If collectivist values are sufficiently strong, postings reduce performance.
4. The performance gap between the best performer and the median performer displays a negative complementarity:

$$\frac{\partial^2}{\partial \lambda \partial \rho} \left(\max_i y_i - \text{median}_i(y_i) \right) < 0. \quad (6)$$

Interpretation: The model rationalizes why named postings can backfire in Phase 1 sites: the strongest drivers reduce performance to avoid publicly distancing themselves from teammates.

5 Empirical specifications

5.1 Pooled intent-to-treat effect

A pooled treatment regression can be written as:

$$DRIVE_{it} = \alpha_1(Post_{it} \times T1_i) + \alpha_2(Post_{it} \times T2_i) + D'_{it}\beta + \varepsilon_{it}, \quad (7)$$

where $T1_i$ denotes named posting, $T2_i$ denotes IDed posting, and D_{it} includes controls and fixed effects.

Interpretation: This specification asks whether postings have an average effect when Phase 1 and pre-Phase 1 sites are pooled together. The answer is close to no.

5.2 Triple-difference specification

The main empirical model is:

$$DRIVE_{it} = \alpha_1(Post_{it} \times T1_i \times Phase1_i) + \alpha_2(Post_{it} \times T2_i \times Phase1_i) \quad (8)$$

$$+ D'_{it}\beta + \varepsilon_{it}. \quad (9)$$

The vector D_{it} includes all lower-order terms and, depending on the specification, site controls, calendar-date fixed effects, driver fixed effects, Phase 1 manager fixed effects, regional fixed effects, and route characteristics.

Interpretation: The coefficient α_1 is the key difference-in-difference-in-differences estimate. A negative α_1 means that named posting performs worse in Phase 1 sites than in pre-Phase 1 sites.

5.3 Selection diagnostics

Because Phase 1 is not researcher-randomized, the authors conduct several checks:

- balance tests between Phase 1 and pre-Phase 1 sites;
- Oster/Altonji-style proportional selection analysis;
- propensity-score matched sample analysis;
- Heckman correction in supplemental analysis;
- compliance-based robustness checks.

Interpretation: These checks ask whether site selection into Phase 1 can explain the negative interaction between Phase 1 and named postings. The paper argues that selection would have to be implausibly strong and in the wrong direction.

5.4 Mechanism tests

The paper studies mechanisms using:

- named versus IDed postings;
- heterogeneity by pre-posting performance quartile;
- engagement survey responses measuring collectivist orientation;
- an instrumental/placebo engagement index based on individual satisfaction with compensation and benefits.

Interpretation: The mechanism tests are designed to distinguish a collectivist-values explanation from generic dissatisfaction, low morale, or measurement artifacts.

6 Identification and empirical design

6.1 What is randomized

- The performance posting condition is randomized.
- Randomization is stratified by EOBR rollout week and Phase 1 status.
- This supports causal interpretation of the posting treatment conditional on Phase 1 strata.

Interpretation: The experiment directly identifies the effect of postings within the site environment into which the posting is introduced.

6.2 What is not randomized

- Phase 1 is not randomized by the researchers.
- The firm rolls out Phase 1 over time according to operational implementation constraints.
- Phase 1 sites are larger in the full sample and differ on some observables, although matching reduces imbalance.

Interpretation: The paper’s most demanding identifying assumption is that unobserved Phase 1 selection does not drive the differential response to named postings.

6.3 Why the design is informative

- The postings are randomly assigned.
- The outcomes are objective performance data from trucks.
- The firm does not change driver pay incentives during Phase 1.
- Phase 1 is values-oriented rather than tools-oriented during the experimental window.
- Named and IDed postings create a mechanism contrast.

Interpretation: The design isolates a rare interaction between a randomized formal practice and a naturally staggered values intervention.

6.4 Limitations

- Phase 1 is quasi-random rather than fully randomized.
- The setting is one firm in one industry.
- Compliance with postings is imperfect across sites.
- The engagement survey has limited response rates.
- The experiment studies a short-run response during technology rollout.

Interpretation: The paper provides strong evidence of contingency within this setting but does not claim that all collectivist cultures will reject all forms of performance transparency.

7 Tables and figures

7.1 Figure 1: experimental timeline

Read:

- EOBR installation begins in late August 2013 and proceeds weekly across sites.
- Posting begins six weeks after each site's EOBR installation.
- Driver performance data are collected from September 2013 to April 2014.
- Phase 1 rollout starts much earlier and continues much longer than the posting experiment.

Economic message: The experiment overlays a fast randomized posting rollout on a slow values-transformation rollout.

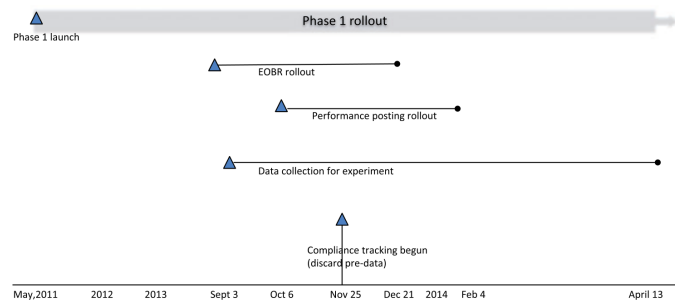


FIGURE 1
Timeline of experiment

7.2 Table 1 and Table 2: balance

Read:

- Table 1 shows balance between control, named posting, and IDed posting groups.
- Treatment arms are broadly balanced on site characteristics and pre-treatment performance.
- Table 2 compares Phase 1 and pre-Phase 1 sites.
- The full sample has imbalances, especially in site size, but the matched sample reduces those differences substantially.

Economic message: Posting assignment is experimentally credible, while Phase 1 status requires additional selection checks.

TABLE 1
Balance between control and treatment groups

	Control Mean	Treat-ment 1 (names) Mean	Treat-ment 2 (IDs) Mean	Diff (T1-C) <i>p</i> -value	Diff (T2-C) <i>p</i> -value	Diff (T2-T1) <i>p</i> -value
A. Full sample						
Site characteristics						
No. of sites	47	50	45	n/a	n/a	n/a
Phase 1 status	0.30	0.26	0.47	0.681	0.098	0.036
Tractors/site	25.00	25.32	23.73	0.924	0.664	0.624
Distance/trip	123.57	129.99	130.42	0.308	0.259	0.947
Eastern region	0.44	0.44	0.30	0.966	0.173	0.176
Central region	0.33	0.22	0.39	0.220	0.551	0.067
Western region	0.22	0.34	0.30	0.207	0.398	0.702
Pre-treatment driver performance						
Miles per gallon	6.76	6.87	6.82	0.254	0.573	0.537
Gap score	2.03	1.98	1.85	0.756	0.307	0.441
Shift score	91.28	91.23	91.80	0.927	0.445	0.435
Excess idle time	0.10	0.10	0.11	0.979	0.344	0.381
Fuel lost	0.31	0.32	0.29	0.596	0.276	0.111
B. Matched sample						
Site characteristics						
No. of sites	25	26	31	n/a	n/a	n/a
Phase 1 status	0.52	0.38	0.58	0.341	0.657	0.145
Tractors/site	28.64	28.15	24.16	0.909	0.226	0.275
Distance/trip	127.53	127.56	128.34	0.993	0.914	0.921
Eastern region	0.39	0.42	0.30	0.826	0.496	0.347
Central region	0.49	0.31	0.37	0.230	0.424	0.649
Western region	0.13	0.30	0.33	0.238	0.092	0.611
Pre-treatment driver performance						
Miles per gallon	6.65	6.76	6.74	0.348	0.392	0.790
Gap score	1.90	1.95	1.87	0.770	0.868	0.669
Shift score	91.99	91.37	92.28	0.368	0.672	0.246
Excess idle time	0.10	0.12	0.12	0.380	0.349	0.816
Fuel lost	0.31	0.32	0.30	0.617	0.644	0.421

Total number of sites masked for confidentiality purposes. *Phase 1 status* is an indicator variable that equals one if site has launched the first phase of its business transformation initiative after 2011, the initial year, and at least three months before the commencement of the experiment. *Tractors/site* is the number of tractors assigned to a given site, a measure of the size of the establishment. *Distance/trip* is the average number of miles per trip, a measure of the nature of the route (shorter routes typically denote more urban settings). *Miles per gallon* is the average actual miles per gallon for a

Table 1: balance between control and treatment groups

TABLE 2
Balance between Phase 1 and Pre-Phase 1 sites

	Full sample			Matched sample		
	Pre-Phase 1 Mean	Phase 1 Mean	Diff <i>p</i> -value	Pre-Phase 1 Mean	Phase 1 Mean	Diff <i>p</i> -value
Site characteristics						
No. of sites	94	48	n/a	41	41	n/a
Tractors/site	20.35	33.25	0.000	25.00	28.59	0.251
Distance/trip	128.82	126.39	0.651	127.88	127.82	0.993
Eastern region	0.27	0.39	0.155	0.33	0.41	0.438
Central region	0.41	0.37	0.626	0.40	0.36	0.712
Western region	0.32	0.24	0.357	0.28	0.23	0.656
Control group	0.35	0.29	0.480	0.29	0.32	0.813
Treatment group 1	0.39	0.27	0.149	0.39	0.24	0.158
Treatment group 2	0.26	0.44	0.027	0.32	0.44	0.260
Pre-treatment driver performance						
Miles per gallon	6.88	6.71	0.039	6.69	6.75	0.492
Gap score	1.98	1.92	0.647	1.89	1.92	0.855
Shift score	91.19	91.89	0.222	91.98	91.84	0.813
Excess idle time	0.10	0.11	0.581	0.12	0.11	0.309
Fuel lost	0.31	0.30	0.541	0.30	0.31	0.704

See Table 1 caption for variable definitions. The number of Phase 1 sites is approximately half of the number of Pre-Phase 1 sites.

Table 2: balance between Phase 1 and pre-Phase 1 sites

7.3 Figure 2 and Figure 3: preliminary event-study evidence

Read:

- Figure 2 pools Phase 1 and pre-Phase 1 sites and shows little average posting effect.
- Figure 3 separates Phase 1 from pre-Phase 1 sites.
- Named postings improve performance in pre-Phase 1 sites but reduce performance in Phase 1 sites.
- IDed postings show much weaker effects.

Economic message: The average effect masks a sharp contingency: performance posting works only when it is culturally aligned.

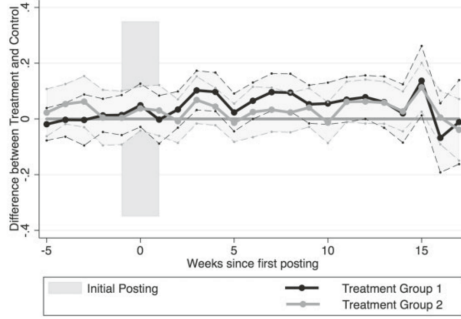


FIGURE 2
Impact of postings on driver performance

See caption to Table 1 for definitions. Error bars reflect 90% confidence intervals, clustered by site. See [Supplementary Figure A5](#) for disaggregated outcomes.

Figure 2: pooled effect of postings

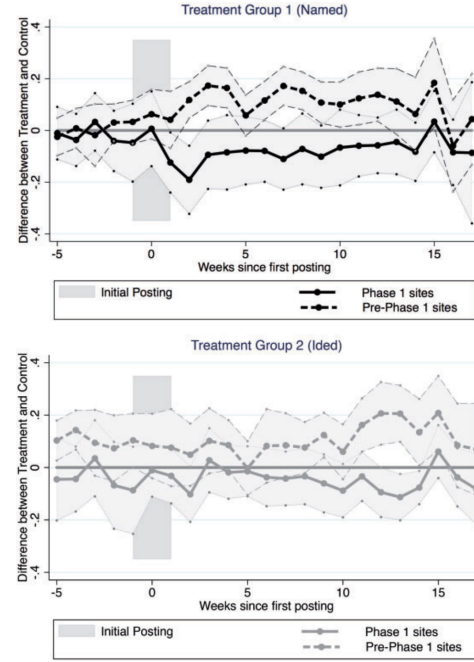


FIGURE 3
Impact of postings

Figure 3: effect by Phase 1 status

7.4 Table 3 and Table 4: main intent-to-treat estimates

Read:

- Table 3 shows no significant pooled effect of postings on all sites.
- Table 4 shows the main triple-difference result.
- The coefficient on $Post \times Treatment\ Group\ 1 \times Phase1$ is negative and statistically significant across specifications.
- The coefficient on the same interaction for Treatment Group 2 is small and insignificant.
- Named postings help pre-Phase 1 sites but hurt Phase 1 sites.

Economic message: The performance consequences of public rankings depend on whether the worksite has adopted collectivist values.

Dependent variables	(1)	Driving efficiency index (2)	(3)	(4)
Post*Treatment group 1	0.0219 (0.0288)	0.0243 (0.0264)	0.0369 (0.0279)	0.0330 (0.0272)
Post*Treatment group 2	-0.0460 (0.0296)	-0.0526* (0.0284)	-0.0306 (0.0265)	-0.0294 (0.0264)
Post	0.0097 (0.0342)	0.0063 (0.0279)	0.0174 (0.0176)	0.0137 (0.0177)
Treatment group 1 (Named)	0.0116 (0.0431)	0.0121 (0.0351)		
Treatment group 2 (IDed)	0.0489 (0.0439)	-0.0021 (0.0467)		
Constant	0.0344 (0.0295)	0.0217 (0.1324)	0.0641*** (0.0143)	0.0089 (0.0572)
Controls	N	Y	N	Y
Driver fixed effects	N	N	Y	Y
Observations	285,606	285,606	285,606	285,606
No. of drivers	5,001	5,001	5,001	5,001
No. of sites (clusters)	142	142	142	142
Adjusted R ²	0.005	0.056	0.498	0.508

For all dependent variables, lower scores represent better performance (more efficient driving). See Table 1 for variable definitions. Controls include *Tractors/site*, *Distance/trip*, *potential MPG*, and fixed effects for the Phase 1 regional manager (the manager in charge of overseeing the Phase 1 implementation), region, and calendar date. Standard errors clustered by site. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 3: pooled posting effects

Dependent variables	(1)	Driving efficiency index (2)	(3)	(4)
Post*Treatment group 1*Phase 1	-0.1258** (0.0509)	-0.1164** (0.0492)	-0.1117** (0.0528)	-0.1032** (0.0522)
Post*Treatment group 2*Phase 1	0.0009 (0.0598)	-0.0028 (0.0577)	-0.0034 (0.0545)	-0.0098 (0.0541)
Post*Treatment group 1	0.0658* (0.0354)	0.0650** (0.0319)	0.0752** (0.0331)	0.0685** (0.0326)
Post*Treatment group 2	-0.0467 (0.0499)	-0.0530 (0.0472)	-0.0298 (0.0414)	-0.0250 (0.0408)
Post*Phase 1	0.0100 (0.0355)	0.0145 (0.0348)	0.0056 (0.0358)	0.0063 (0.0359)
Treatment group 1*Phase 1	-0.1152 (0.0816)	-0.0345 (0.0847)		
Treatment group 2*Phase 1	-0.0976 (0.0837)	-0.0670 (0.0882)		
Post	0.0125 (0.0380)	-0.0005 (0.0299)	0.0150 (0.0233)	0.0109 (0.0234)
Phase 1	-0.0024 (0.0520)	-0.0117 (0.0591)		
Treatment group 1 (Named)	0.0514 (0.0557)	0.0190 (0.0449)		
Treatment group 2 (IDed)	0.1057* (0.0556)	0.0407 (0.0578)		
Constant	0.0372 (0.0419)	-0.0048 (0.1393)	0.0638*** (0.0137)	0.0109 (0.0571)
Controls	N	Y	N	Y
Driver fixed effects	N	N	Y	Y
Observations	285,606	285,606	285,606	285,606
# Drivers	5,001	5,001	5,001	5,001
# Sites	142	142	142	142
Adjusted R ²	0.011	0.058	0.498	0.508

For all dependent variables, lower scores represent better performance (more efficient driving). See Table 1 for variable definitions and Table 3 for list of controls. Errors clustered by site. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 4: triple-difference effects

7.5 Table 5 and Table 6: selection robustness

Read:

- Table 5 reports proportional-selection diagnostics in the spirit of Oster and Altonji et al.
- The implied amount of unobserved selection required to overturn the result is beyond conventional plausibility bounds.
- Table 6 repeats the main analysis on a matched sample.
- The negative interaction for named posting and Phase 1 becomes larger in magnitude in the matched analysis.

Economic message: The results are not easily explained by non-random assignment of Phase 1 sites.

	Log(Gap Score)	
$\Pi =$	1.3	2.0
δ	4.1894	2.0290
β	-0.1018	-0.0678
β_{hat}	-0.1164	-0.1164

Displays output statistics from Oster (2014) proportional selection bias analysis using model specified in Table 4 Column (2). The analysis examines the stability of the coefficient estimate on Post * Treatment Group 1 * Phase 1.

Table 5: proportional selection bias analysis

Dependent variables	(1)	Driving efficiency index (2)	(3)	(4)
Post*Treatment group 1*Phase 1	-0.1969*** (0.0644)	-0.1831*** (0.0635)	-0.1616** (0.0666)	-0.1506** (0.0663)
Post*Treatment group 2*Phase 1	-0.0474 (0.0698)	0.0462 (0.0680)	0.0456 (0.0634)	0.0362 (0.0638)
Post*Treatment group 1	0.1012** (0.0479)	0.0917* (0.0471)	0.0859* (0.0467)	0.0791* (0.0459)
Post*Treatment group 2	-0.0977 (0.0632)	-0.0986 (0.0612)	-0.0872 (0.0529)	-0.0800 (0.0527)
Post*Phase 1	0.0463 (0.0428)	0.0409 (0.0417)	0.0278 (0.0427)	0.0285 (0.0435)
Treatment group 1*Phase 1	-0.0737 (0.0897)	-0.0721 (0.1033)		
Treatment group 2*Phase 1	-0.0360 (0.1108)	-0.1614 (0.0980)		
Post	-0.0031 (0.0515)	0.0162 (0.0400)	0.0116 (0.0350)	0.0060 (0.0354)
Phase 1	-0.0594 (0.0658)	-0.0475 (0.0717)		
Treatment group 1 (Named)	0.0386 (0.0662)	0.0014 (0.0708)		
Treatment group 2 (IDed)	0.0361 (0.0771)	0.1678** (0.0789)		
Constant	0.0784 (0.0559)	0.0754 (0.1777)	0.0574*** (0.0174)	-0.0244 (0.0755)
Controls	N	Y	N	Y
Driver fixed effects	N	N	Y	Y
Observations	181,404	181,404	181,404	181,404
# Drivers	3,127	3,127	3,127	3,127
# Sites	82	82	82	82
Adjusted R ²	0.014	0.060	0.500	0.509

See Table 1 for variable definitions and Table 3 for list of controls. Errors clustered by site. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 6: matched sample analysis

7.6 Table 7: heterogeneity by pre-posting performance quartile

Read:

- Table 7 splits drivers into quartiles based on pre-posting driving efficiency.
- The negative Phase 1 interaction for named postings is strongest among the top and bottom performance quartiles.
- The top-quartile effect is consistent with high performers reducing effort to avoid social friction.
- The bottom-quartile effect is consistent with shame or disengagement when poor performance is publicly identifiable.

Economic message: The mechanism is social rather than purely informational: identifiability changes how different types of drivers react.

TABLE 7
Effect by performance quartile

Dependent variables Quartile	Driving efficiency index			
	First (top) (1)	Second (2)	Third (3)	Fourth (Bottom) (4)
Post*Treatment group 1*Phase 1	−0.1431** (0.0646)	−0.0402 (0.0605)	−0.0163 (0.0707)	−0.2299*** (0.0824)
Post*Treatment group 2*Phase 1	0.0386 (0.0706)	−0.0027 (0.0622)	0.0048 (0.0698)	−0.0394 (0.0911)
Post*Treatment group 1	0.0788* (0.0469)	0.0312 (0.0404)	0.0738* (0.0424)	0.0829* (0.0481)
Post*Treatment group 2	−0.0297 (0.0588)	−0.0512 (0.0450)	0.0024 (0.0552)	−0.1083* (0.0599)
Post*Phase 1	0.0422 (0.0461)	−0.0021 (0.0434)	−0.0103 (0.0430)	0.0101 (0.0532)
Treatment group 1*Phase 1	−0.0356 (0.0697)	−0.0383 (0.0834)	0.0279 (0.0994)	−0.1297 (0.1290)
Treatment group 2*Phase 1	−0.0079 (0.0816)	−0.0919 (0.0872)	−0.0604 (0.1130)	−0.1863 (0.1369)
Post	−0.1325*** (0.0402)	−0.0427 (0.0368)	0.0290 (0.0345)	0.1349*** (0.0446)
Phase 1	−0.0635 (0.0573)	−0.0153 (0.0594)	0.0181 (0.0717)	0.1006 (0.0887)
Treatment group 1 (names)	0.0217 (0.0484)	0.0592 (0.0490)	−0.0091 (0.0530)	0.0560 (0.0671)
Treatment group 2 (IDs)	0.0175 (0.0626)	0.0821 (0.0566)	0.0588 (0.0633)	0.1021 (0.0921)
Constant	0.4727*** (0.1258)	0.1766 (0.1368)	0.1258 (0.1871)	−0.2763 (0.2020)
Controls	Y	Y	Y	Y
Observations	68,717	69,192	70,719	70,362
# Drivers	1178	1178	1176	1180
# Sites	142	142	142	142
Adjusted R^2	0.087	0.085	0.083	0.107

Quartiles based on average driving efficiency prior to performance postings. Errors clustered by site. ***, **, and * indicate significance at 1%, 5%, and 10%.

7.7 Table 8 and Figure 4: engagement survey mechanism

Read:

- Table 8 replaces Phase 1 status with a survey-based collective index and an instrumental/placebo index.
- The interaction between named postings and the collective index is negative and significant.
- The instrumental/placebo index does not reproduce the same pattern.
- Figure 4 shows that higher collective-index deciles are associated with much lower response to postings, while the placebo index does not show the same negative pattern.

Economic message: The evidence supports the interpretation that collectivist values, not generic satisfaction, are what moderate the effectiveness of postings.

TABLE 8
Effect of ranking and engagement on driver performance

Dependent variables	(1)	Driving efficiency index (2)	(3)	(4)
Post*Treatment Group 1*Collective Index	-0.2436** (0.1008)	-0.2505** (0.0984)	-0.1724** (0.0716)	-0.1892** (0.0710)
Post*Treatment Group 2*Collective Index	-0.0429 (0.0846)	-0.0618 (0.0807)	-0.0355 (0.0608)	-0.0469 (0.0576)
Post*Treatment Group 1*Instrumental Index	0.1027 (0.0765)	0.1015 (0.0682)	0.0373 (0.0493)	0.0492 (0.0488)
Post*Treatment Group 1*Instrumental Index	0.0567 (0.0726)	0.0685 (0.0687)	0.0371 (0.0592)	0.0379 (0.0552)
Post*Treatment Group 1	0.3957** (0.1910)	0.4023** (0.1921)	0.4129** (0.1537)	0.4316*** (0.1493)
Post*Treatment Group 2	-0.2170 (0.2658)	-0.1852 (0.2554)	-0.1449 (0.1864)	-0.0954 (0.1753)
Post*Collective Index	0.1103 (0.0776)	0.1214 (0.0762)	0.0887* (0.0487)	0.0995** (0.0464)
Post*Instrumental Index	-0.0543 (0.08430)	-0.0712* (0.0370)	-0.0337 (0.0336)	-0.0410 (0.0334)
Treatment Group 1*Collective Index	-0.1265 (0.1004)	-0.1361 (0.1048)		
Treatment Group 2*Collective Index	0.0410 (0.0951)	-0.0047 (0.0973)		
Treatment Group 1*Instrumental Index	-0.0053 (0.1127)	-0.0196 (0.0980)		
Treatment Group 2*Instrumental Index	-0.0036 (0.1010)	-0.0043 (0.0919)		
Post	-0.0391 (0.2185)	0.0680 (0.1810)	-0.0944 (0.1343)	-0.1257 (0.1390)
Collective Index	-0.0073 (0.0691)	0.0058 (0.0797)		
Instrumental Index	0.0485 (0.0825)	0.0483 (0.0778)		
Treatment Group 1	0.4687 (0.3048)	0.4785* (0.2506)		
Treatment Group 2	-0.1940 (0.2536)	-0.3313 (0.2929)		
Constant	-0.1732 (0.1815)	0.3330 (0.3424)	-0.0128 (0.0319)	0.0828 (0.1549)
Controls	N	N	N	Y
Fixed effects	N	N	Y	Y
Observations	29,774	29,774	29,774	29,774
# Drivers	486	486	486	486
# Sites (clusters)	43	43	43	43
Adjusted R ²	0.047	0.079	0.540	0.553

Table 8: engagement survey mechanism

EKE ET AL. THE CONTINGENT EFFECT OF MANAGEMENT PRACTICES 14

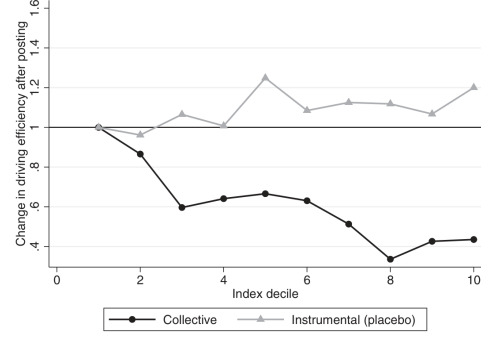


FIGURE 4
Posting response by engagement survey response

z, relative to control group, in driving efficiency between pre- and post-periods, by decile of survey response questions. Deciles: two indices of survey questions constructed to capture, respectively, collective identification (or team affiliation), and instrumental action, such as satisfaction with pay and benefits.

Figure 4: posting response by survey-response decile

8 Short summary

- The paper studies whether management practices have contingent effects depending on organizational values.
- A transportation company installs EOBR technology, making individual driver performance observable.
- Sites are randomly assigned to control, named public postings, or IDed anonymous postings.
- The firm is simultaneously rolling out Phase 1 of a lean initiative focused on teamwork and empowerment.
- Named postings improve performance in sites not yet exposed to Phase 1.
- Named postings reduce performance in Phase 1 sites.
- IDed postings have much weaker effects, suggesting identifiability and social comparison are central.
- Balance, matching, proportional-selection, and Heckman-style checks support the robustness of the result.
- Survey evidence shows that collectivist orientation, not generic job satisfaction, explains the pattern.
- The paper's main lesson is that formal management practices and intangible organizational values must be jointly designed.

9 Conclusion

9.1 Core conclusion

Blader, Gartenberg, and Prat show that the same management practice can have opposite effects depending on the values environment in which it is introduced. Publicly posting employee performance can improve performance in individualistic sites but reduce performance in sites that have been shifted toward teamwork and empowerment.

9.2 Economic intuition

Named performance postings create competition and status comparison. In sites that have not internalized collectivist values, this can motivate drivers to outperform peers. In Phase 1 sites, however, the same public ranking conflicts with team-oriented values. High performers may reduce effort to avoid embarrassing coworkers, while low performers may disengage when publicly identified.

9.3 Takeaway

The paper's broader implication is that managerial practices are not plug-and-play. Managers and researchers must evaluate practices as part of a system that includes both formal tools and intangible values. A practice that is productive in one culture can be counterproductive in another.